

CLASS 11TH BIOLOGY (SECTION B: ZOOLOGY) ? CRUCIAL YEAR QUESTIONS

ANIMAL KINGDOM & STRUCTURAL ORGANISATION

--- Long Answer Type Questions ---

- Q49.** Write down the salient features of phylum Coelenterata, Platyhelminthes, Echinodermata, and Annelida.
- Q50.** How are urochordates different from cephalochordates? Describe the differences between chordates and non-chordates.
- Q51.** All vertebrates are chordates but all chordates are not vertebrates. Justify the statement with examples.
- Q52.** Describe the respiratory, circulatory, and reproductive system of Frog.

--- Short Answer Type Questions ---

- Q53.** What is a Coelom? Give one example each of a Pseudocoelomate and Acoelomate.
- Q54.** Differentiate between diploblastic and triploblastic animals with suitable examples.
- Q55.** What is the importance of the canal system in sponges?
- Q56.** Mention two differences between open and closed circulatory systems.
- Q57.** What is a water vascular system? Name one phylum where it is found.
- Q58.** Explain the importance of coelom in animals.
- Q59.** List three salient features of class Reptilia.

--- Very Short Answer Type Questions ---

- Q60.** What is metamerism? Give one phylum where it is observed.
- Q61.** Which germ layer gives rise to the lining of the digestive tract?
- Q62.** Write two functions of the canal system in sponges.
- Q63.** State two differences between polyp and medusa forms.
- Q64.** Mention two adaptations in aves for flight.
- Q65.** Mention two unique features of mammals.

HUMAN PHYSIOLOGY

--- Long Answer Type Questions ---

- Q66.** Describe the structure and function of the human respiratory system.
- Q67.** Explain the process of blood circulation in humans.
- Q68.** What are contractile proteins? Give a diagrammatic representation of thin and thick filaments.
- Q69.** What are fibrous, cartilaginous, and synovial joints? How are they different from each other?
- Q70.** Discuss the mechanism of muscle contraction in the light of the 'sliding filament theory'.

- Q71.** Describe the structure and function of the human excretory system.
- Q72.** Describe the counter-current mechanism.
- Q73.** Explain the structure and function of the human nervous system.
- Q74.** Describe the process of generation, conduction, and transmission of nerve impulses.
- Q75.** Describe the mechanism of hormone action.
- Q76.** Explain the process of blood clotting.

--- Short Answer Type Questions ---

- Q77.** Explain two ways by which carbon dioxide is transported in blood.
- Q78.** Define inspiratory reserve volume, expiratory reserve volume, and vital capacity of lungs.
- Q79.** What is the oxygen dissociation curve? State its significance.
- Q80.** What is the Bohr effect? Why is it important for respiration?
- Q81.** What happens to the respiratory process in a person going up a hill?
- Q82.** How is the process of respiration regulated?
- Q83.** Give a brief account of the cardiac cycle.
- Q84.** Write three differences between open and closed circulatory systems.
- Q85.** Write a short note on:
- (I) Angina
 - (II) Occupational respiratory disorders
 - (III) Micturition
 - (IV) Osmoregulation
 - (V) Myasthenia Gravis
 - (VI) Osteoporosis
- Q86.** Describe the role of liver, lungs, and skin in excretion.
- Q87.** Distinguish between skeletal muscle and cardiac muscle.
- Q88.** Give a brief account of the mechanism of synaptic transmission.
- Q89.** Differentiate between:
- (I) Afferent and efferent neurons
 - (II) Myelinated and unmyelinated axons
 - (III) Cranial nerves and spinal nerves
 - (IV) Cerebrum and Cerebellum
- Q90.** Write a short note on the functions of the following hormones:
- (I) Thyroid hormones
 - (II) Insulin and Glucagon
- Q91.** Give a brief account of the Adrenal gland, Pancreas, and Thymus.

CELL, CELL CYCLE & DIVISION

--- Long Answer Type Questions ---

- Q92.** Explain the cell theory and discuss the contributions of scientists who developed it.
- Q93.** Draw a plant cell and explain the structure and function of its major organelles.
- Q94.** Describe the fluid mosaic model of the cell membrane and explain its functions.

- Q95.** Explain the structure of the nucleus and its role in cellular activities.
- Q96.** Discuss the endomembrane system, including the endoplasmic reticulum, Golgi apparatus, lysosomes, and vacuoles.
- Q97.** Explain the structure and functions of mitochondria and plastids.
- Q98.** Describe ribosomes and their role in protein synthesis.
- Q99.** Explain cell division (mitosis and meiosis) and their importance in growth and reproduction.
- Q100.** Discuss the structure and functions of lysosomes, vacuoles, and peroxisomes.
- Q101.** Explain the differences between plant and animal cells with a labeled diagram.
- Q102.** Describe the cell wall and its significance in plant cells.
- Q103.** Explain the cellular organelles involved in energy production and storage.

--- **Short Answer Type Questions** ---

- Q104.** Define cell cycle. Name its phases.
- Q105.** What is interphase? Why is it called a preparatory phase?
- Q106.** Differentiate between G1 and G2 phase.
- Q107.** What happens in the S-phase of the cell cycle?
- Q108.** Mention one significance of the M-phase.
- Q109.** Define quiescent phase (G0 phase).
- Q110.** Why is interphase not called a resting phase?
- Q111.** Write one difference between mitotic and meiotic cell division.
- Q112.** What are mesosomes? Give one function.
- Q113.** Why is the plasma membrane called selectively permeable?
- Q114.** State two functions of the endoplasmic reticulum.
- Q115.** Why is the Golgi apparatus called the 'post office of the cell'?
- Q116.** What are ribosomes? Write their function.
- Q117.** Why are mitochondria called the 'powerhouse of the cell'?
- Q118.** What is the difference between chromatin and chromosomes?
- Q119.** Write two functions of the nucleolus.
- Q120.** Name two cell organelles with double membranes.
- Q121.** What are lysosomes? Why are they called 'suicide bags'?
- Q122.** Define plasmodesmata and state its role.
- Q123.** What are microbodies? Give one example.
- Q124.** Differentiate between cilia and flagella.

BIOMOLECULES

--- **Long Answer Type Questions** ---

- Q125.** Define carbohydrates. Classify them into mono-, di-, and polysaccharides with suitable examples.

- Q126.** Write the structure and role of starch, glycogen, and cellulose.
- Q127.** What are proteins? Describe different levels of protein structure (primary, secondary, tertiary, quaternary) with examples.
- Q128.** Define lipids. Explain the structure of triglycerides and phospholipids.
- Q129.** Distinguish between nucleosides and nucleotides.
- Q130.** What are enzymes? Explain the 'lock and key' hypothesis of enzyme action. Discuss factors affecting enzyme activity (pH, temperature, concentration).

--- Short Answer Type Questions ---

- Q131.** Define biomolecules with examples.
- Q132.** Differentiate between monosaccharides and disaccharides with examples.
- Q133.** Define glycosidic bond.
- Q134.** Write two examples of secondary metabolites.
- Q135.** Distinguish between nucleoside and nucleotide.
- Q136.** Name the nitrogenous bases present only in DNA and only in RNA.
- Q137.** Why are proteins called polymers of amino acids?
- Q138.** Define peptide bond.
- Q139.** Write two properties of enzymes.
- Q140.** What is activation energy?
- Q141.** Name the scientists who proposed the 'double helix' model of DNA.
- Q142.** Differentiate between competitive and non-competitive inhibition of enzymes.
- Q143.** Define phosphodiester bond.
- Q144.** Why are lipids insoluble in water?
- Q145.** Name two essential functions of cholesterol.
- Q146.** Give one role of RNA in protein synthesis.